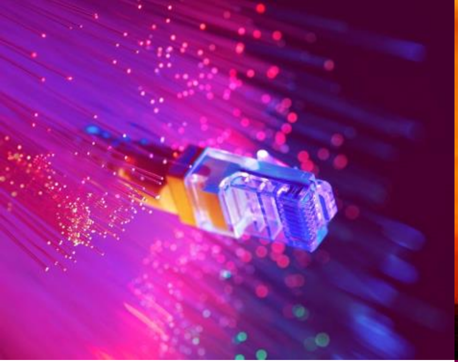
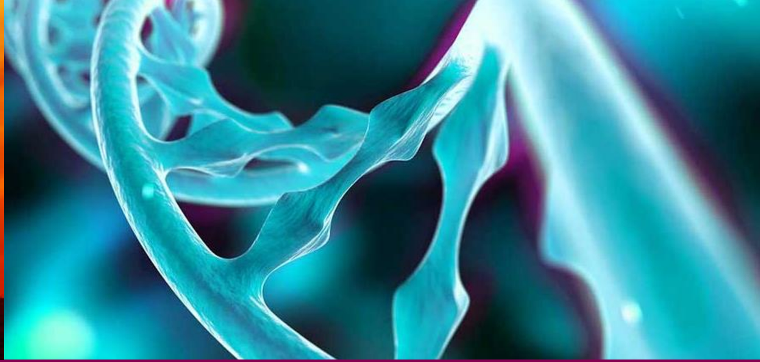
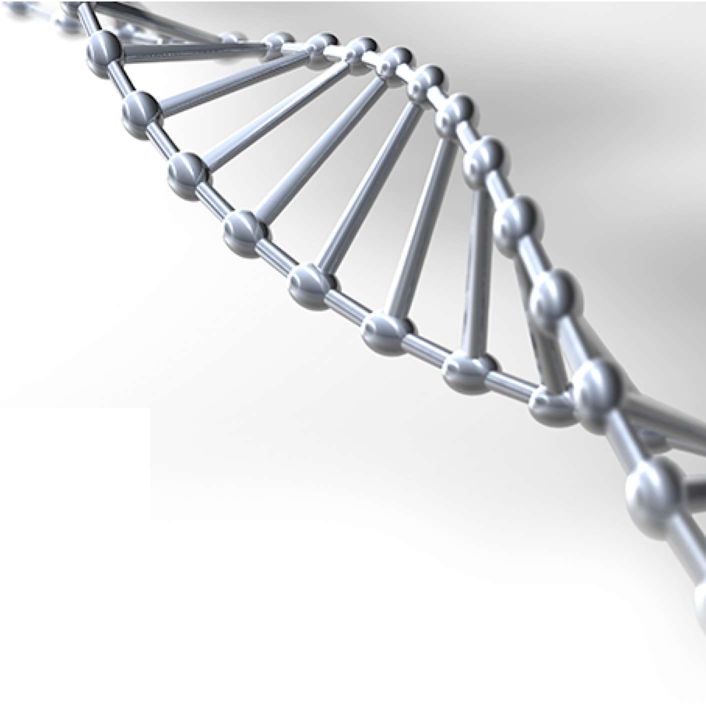




Visium data analysis using Chipster

Spring 2025

Eija Korpelainen, Maria Lehtivaara (Iida Hakulinen)

Introduction

Spatially resolved transcriptomics

Overview

- Intro to spatially resolved transcriptomics
- How does the Visium system work?
- Things to keep in mind when working with Visium data
- What will you learn during this course

Spatially resolved transcriptomics

- Spatial context: Gene expression data overlaid with a tissue image
  - retains organization of tissue and cellular microenvironment
  - cell type identification in the context of heterogeneous tissue
- Several technologies available
  - this course focuses on 10X Genomics Visium data

Visium – how does it work?

- Place tissue slice (frozen or FFPE) on a capture area on a slide
- Capture area contains about 5000 barcoded spots
    - Spot diameter 55 um, center to center distance 100 um
    - NOTE: about 1-10 cells per spot
- Each spot has millions of capture probes
    - 16 nt spatial barcode to track back the location
    - 12 nt UMI for counting
    - 30 nt poly(dT) to capture polyA
- Stain, image, permeabilize cells
- cDNA synthesis
- Library construction

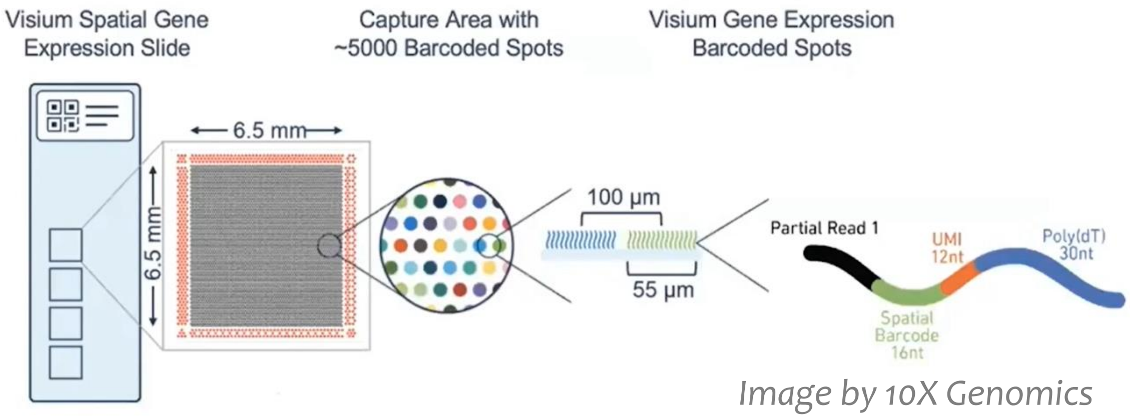


Image by 10X Genomics

Visium data – things to remember

- Each spot typically includes several cells, not just one
- There can be different types of cells in a spot
- The gene expression values measured are an average from the cells in a spot
- Clusters represent a group of spots with **similar composition of cell types**

New Visium HD has higher spatial resolution

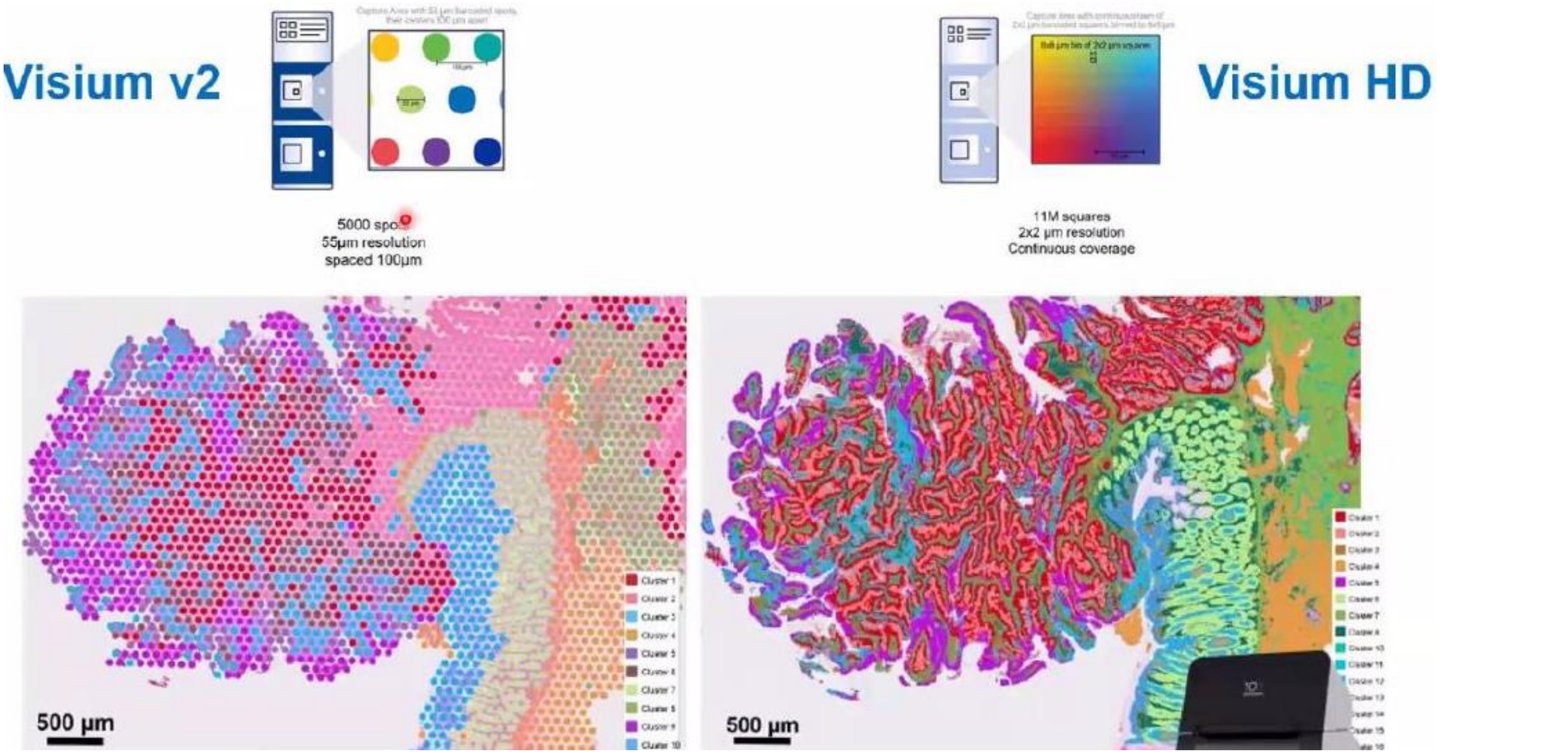
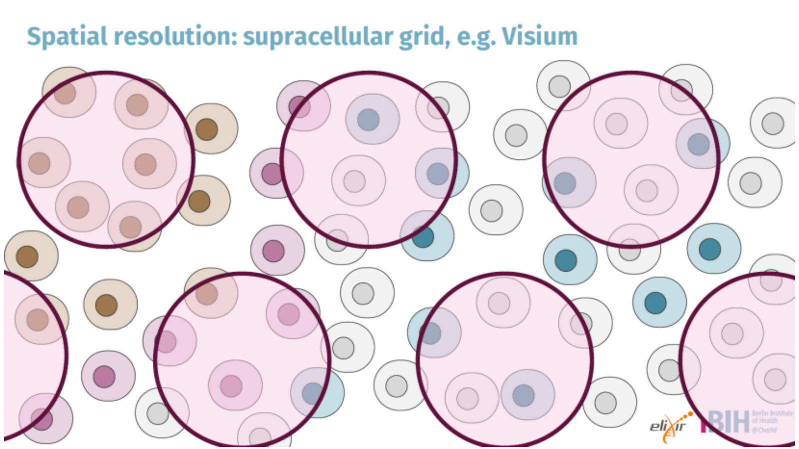
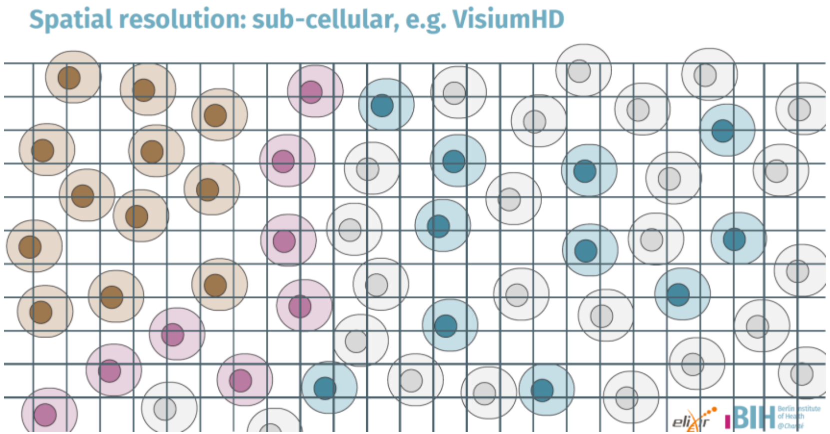


Image by Lars Borm

Visium vs Visium HD: different computational challenge





- Deconvoluting mixed signals in spots

- Aggregating signals into cells
- https://satijalab.org/seurat/articles/visiumhd_analysis_vignette

Image by Naveed Ishaque

During this course you will learn how to

- Create a Seurat object and check the quality of spots
  - Filter out low quality spots (damaged tissue)
- Normalise gene expression values and identify highly variable genes
- Reduce dimensions with PCA using the highly variable genes
- Use the PCs to cluster spots with graph based clustering
  - Visualise clusters (UMAP and overlay with the tissue image)
- Detect spatially variable genes
  - Visualise gene expression on the tissue image
- Subset anatomical regions
- Predict cell type composition in spots: Integrate with scRNA-seq data
- Integrate several samples

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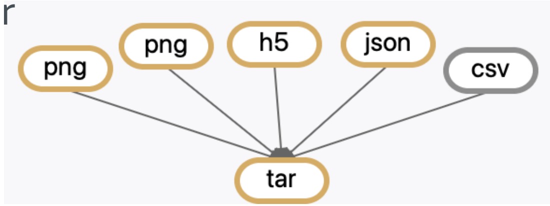
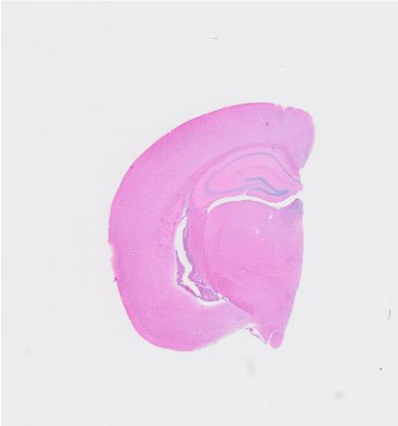
Setting up Seurat object

Spatially resolved transcriptomics

Make a Seurat object with the tool "Seurat – setup and QC"

- Input
  - tar package of 10X Genomics output files from Visium Space Ranger software:
    - More info: https://www.10xgenomics.com/support/software/space-ranger/latest/analysis/outputs/spatial-outputs
    - You can use the Chipster Utilities tool "Make a tar package"
  - Files to be packed
    - **Filtered**_feature_bc_matrix.h5 (= spot by gene expression matrix)
    - Tissue_hires_image.png (= image of the tissue slice)
    - Tissue_lowres_image.png
    - Scalefactors_json.json (= relate the high res image to low res)
    - Tissue_positions_list.csv (check that it doesn't contain the column names) (= spot positions over the image)





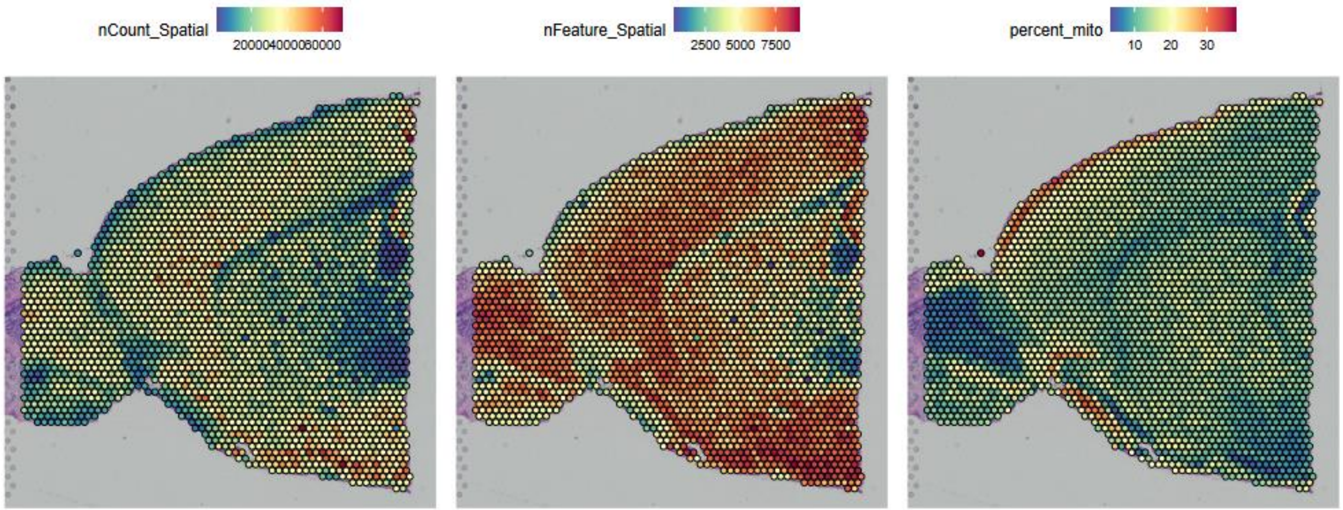
- Output
  - Seurat object containing spot-level expression data and the image of the tissue slice, QC plots

Quality control

Spatially resolved transcriptomics

Violin plot of UMI counts, no of genes, mitochondrial % in spots

QC data plotted on tissue image

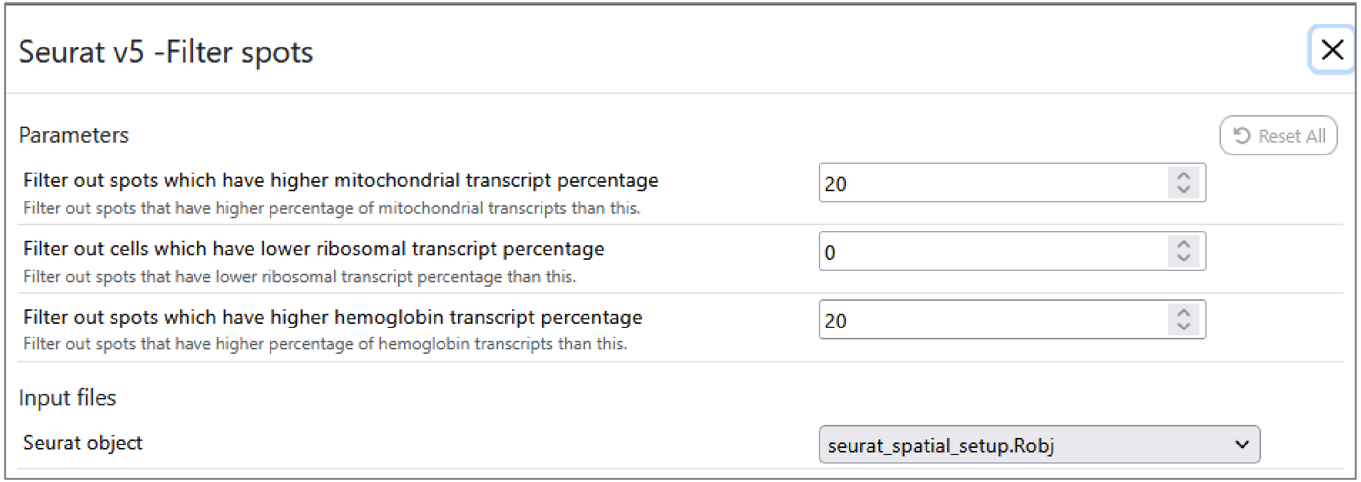


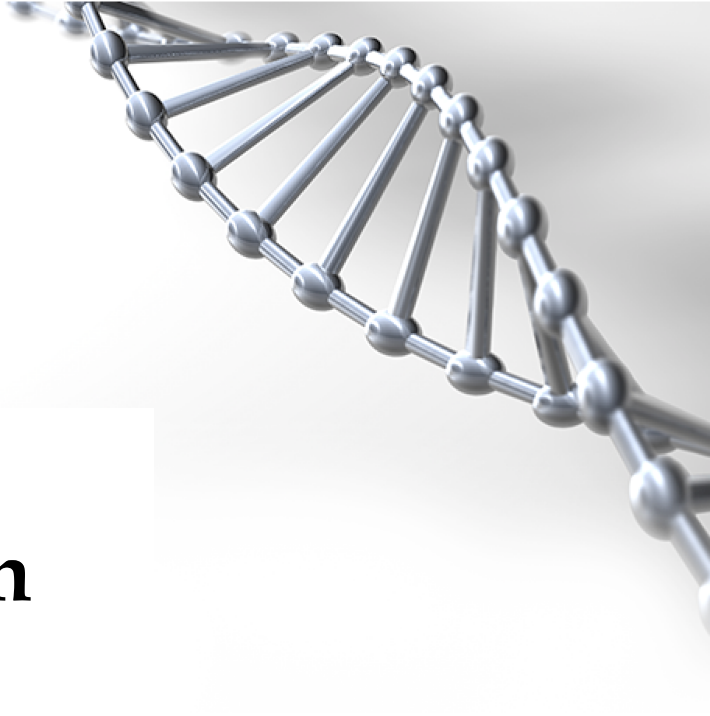
- nCount_Spatial: big differences in UMI counts per spot
  - technical and biological reasons (tissue anatomy) → clever normalization required
- High percentage of mitochondrial reads near the edges, tears and folds of tissue
  - damaged tissue

Filtering

Filtering bad quality spots

- You can filter spots prior to normalization based on
  - Mitochondrial transcript percentage
  - Ribosomal transcript percentage
  - Hemoglobin transcript percentage

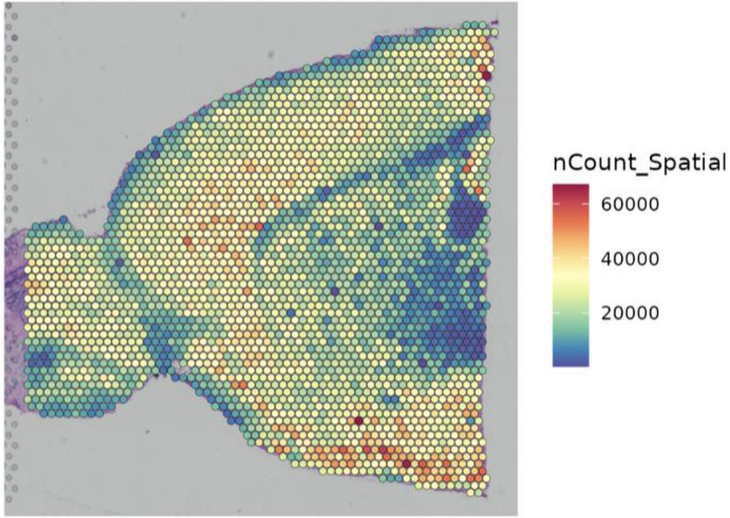




Normalisation

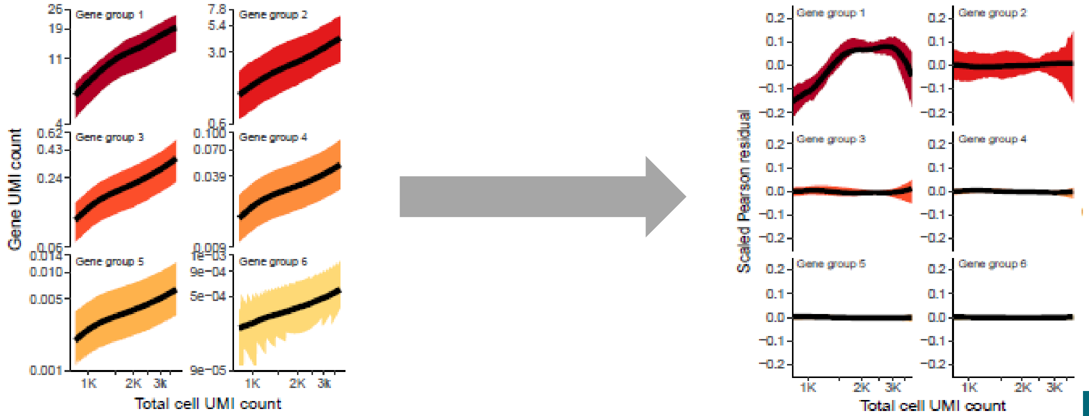
Normalising expression values across spots

- Sequencing depth (number of UMIs per spot) varies significantly between spots
- Normalized expression values of a gene should be independent of sequencing depth
- Variance can be substantial for spatial datasets
- Cell density varies across the tissue → global scaling normalization doesn't work
- Use SCTransform
  - Doesn't force the same "size" on each data point
  - = takes into account the biological variance in cell density



SCTransform

- Models gene expression as a function of sequencing depth using GLM
  - Constrains the model parameters through regularization, by pooling information across genes which are expressed at similar levels
  - Normalized expression values = Pearson residuals from regularized negative binomial regression
- Works well also for high expressing genes
- In addition to normalization, identifies highly variable genes and scales data
- SCTransform v2 even better

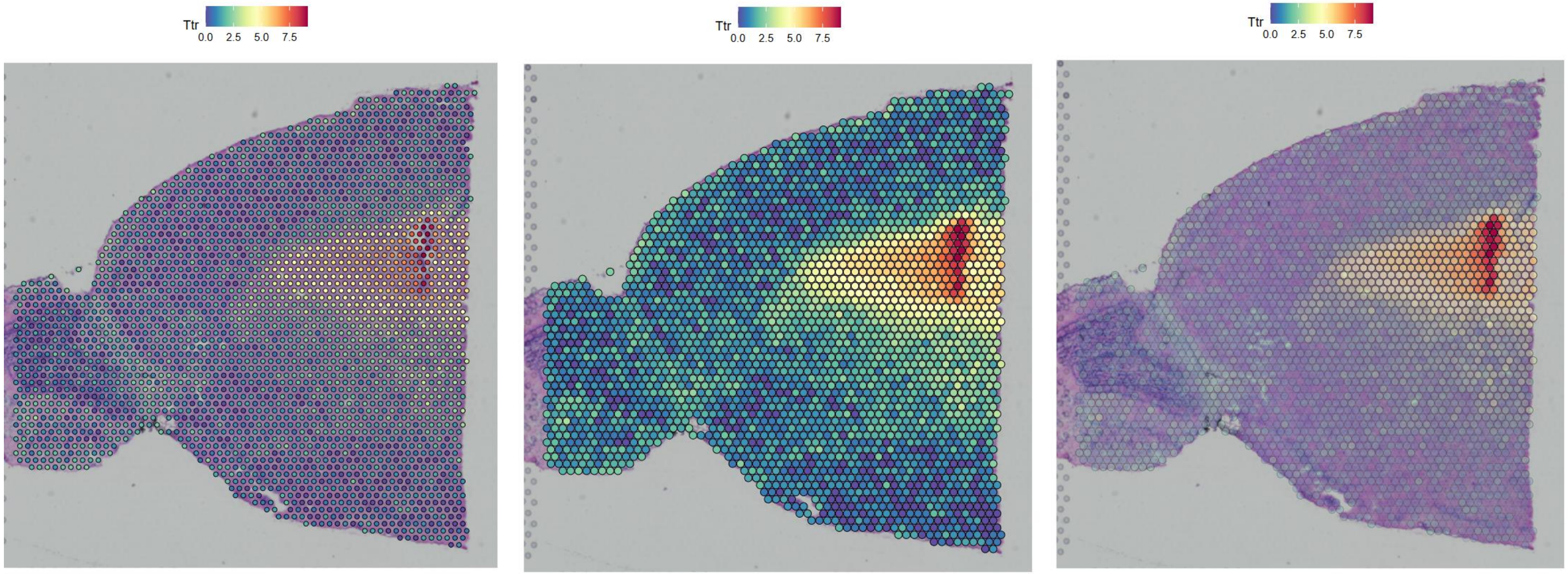


Gene expression visualization

Spatially resolved transcriptomics

Overlay gene expression values on top of histology image

- If you want to see the tissue better you can modify
  - point size
  - transparency of the points (spots with lower expression for gene X are more transparent)



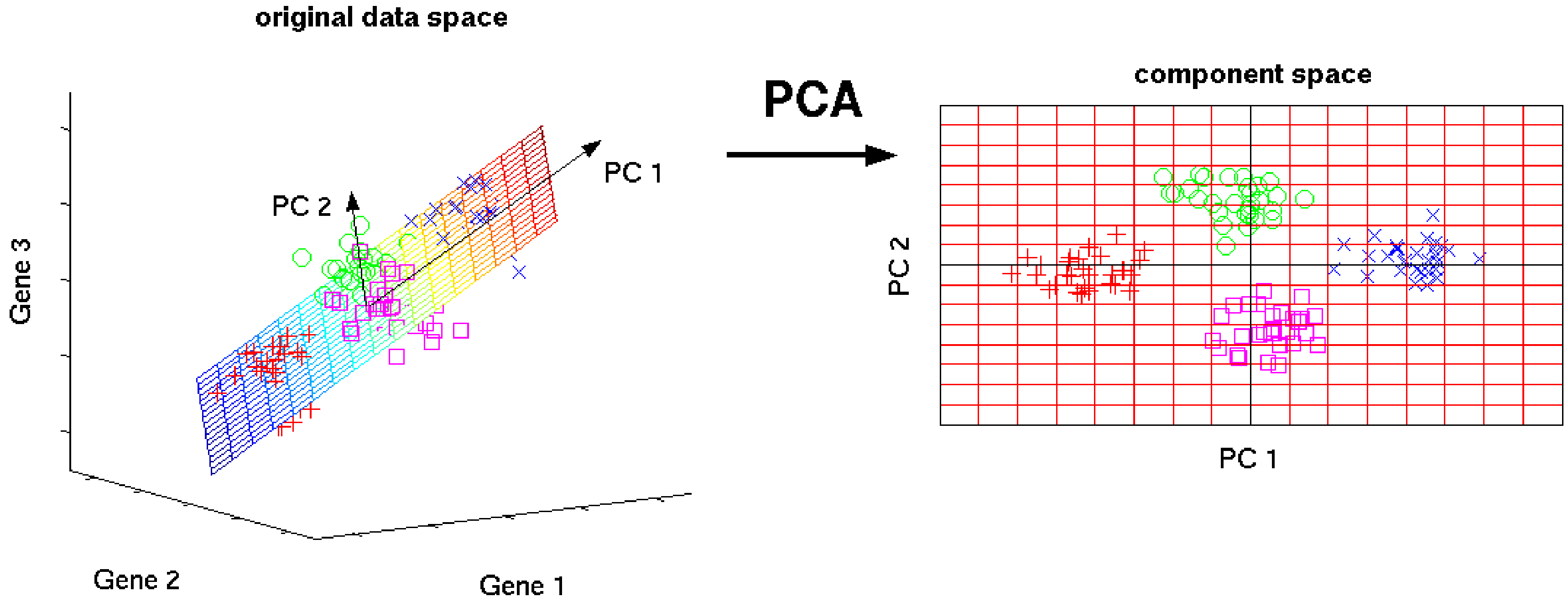
Point size 1

Point size 1.6 (default)

Transparency min 0.1

Dimension reduction

Dimensionality reduction

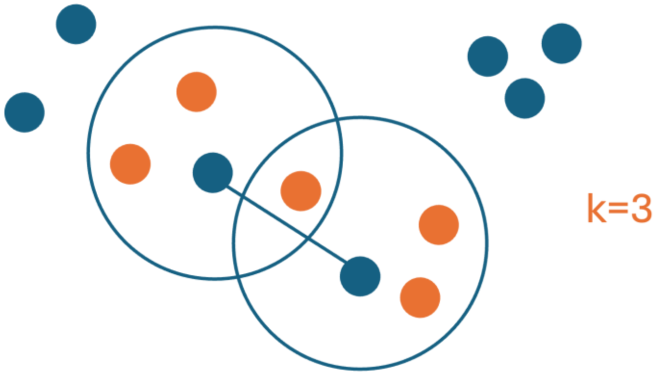


Picture 1. Im, Jonas. "Introduction to PCA" Medium.com, 6th Dec. 2018, https://medium.com/@jamesim2077/introduction-to-pca-principal-component-analysis-c26dffe2a857. Accessed 10 Aug. 2022

Clustering

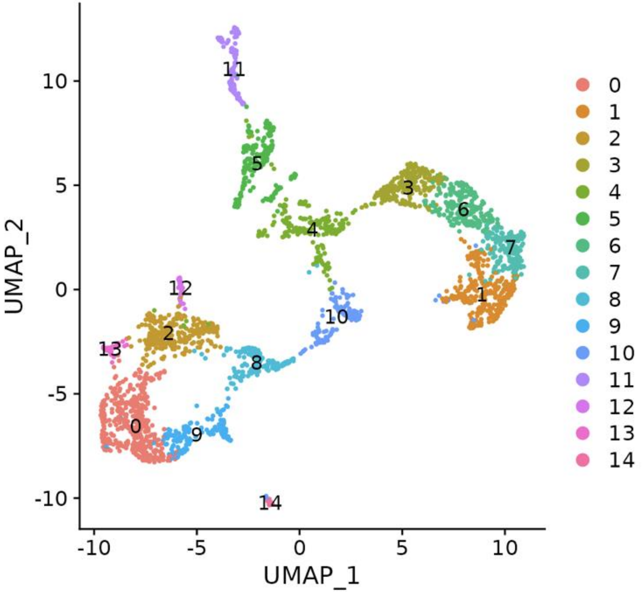
Clustering

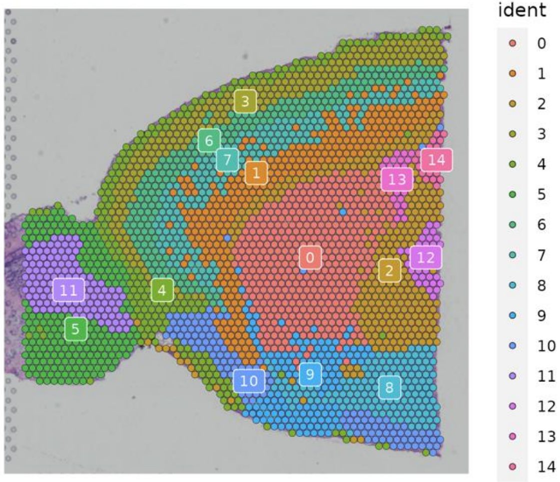
- Divides spots into distinct groups based on gene expression
- We use principal components instead of genes
  - Look at plots from previous step to select the number of PCs
  - Compute clustering in the selected PCA space
- Graph-based clustering approach
  - Shared nearest neighbors
  - Louvain algorithm



Cluster visualisation

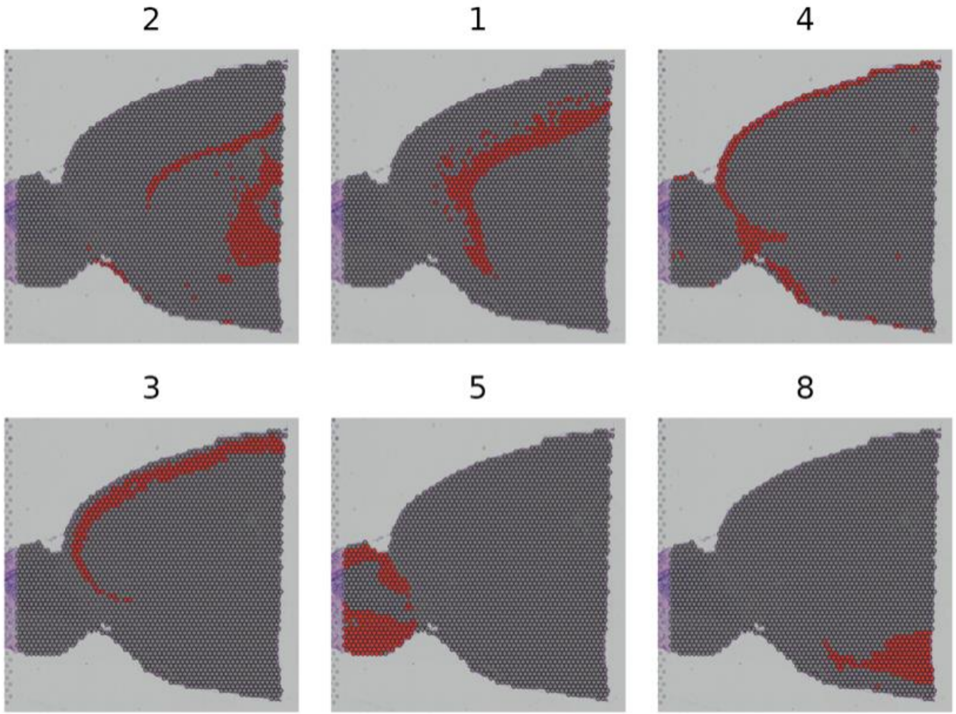
- UMAP plot
- Clusters on top of the slice image





Tricky to see the clusters?

- Using **Visualise clusters** tool and drawing one cluster at a time helps



Identifying spatially variable genes

Identifying spatially variable genes

- Spatially variable gene = gene whose expression level depends on spatial location
- Two ways for identifying spatially variable genes
  - With pre-annotation (cluster information)
  - Without pre-annotation

Identify spatially variable genes based on clusters

- Works when clusters show clear spatial restriction

Seurat v5 -Identify spatially variable genes based on clusters

Parameters

Reset All

First cluster
Cluster you want to identify the differentially expressed for.
1

Second cluster
A second cluster for comparison.
2

Limit testing to genes which are expressed in at least this fraction of spots
Test only genes which are detected in at least this fraction of spots in either cluster. Withholding infrequently expressed genes will speed up testing.
0.01

Limit testing to genes which show at least this fold difference
Test only genes which show on average at least this log2 fold difference between the two groups of spots. Increasing the threshold speeds up testing, but can also miss weaker signals.
0.1

Test for differential expression
wilcox

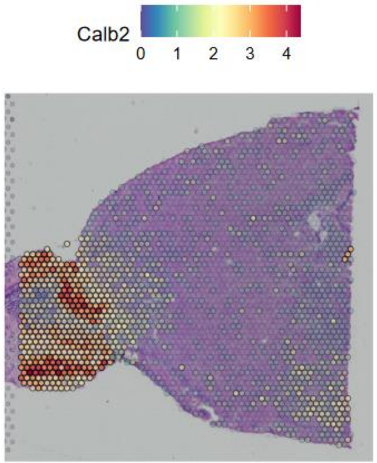
Report only positive marker genes
By default, this tool lists all markers. When this parameter is set to TRUE, only genes with positive log2 fold change are listed in the result file.
FALSE

Number of spatially variable genes to visualize
Choose the number of highest variable genes to visualize.
3

Determine color scale based on all genes
Determine whether the color scale is based on all genes or individual genes. By default, the color scale is determined for each gene individually and may differ between genes.
no

Input files

Seurat object
seurat_obj_clustering.Robj



spatially_variable_genes.tsv •••

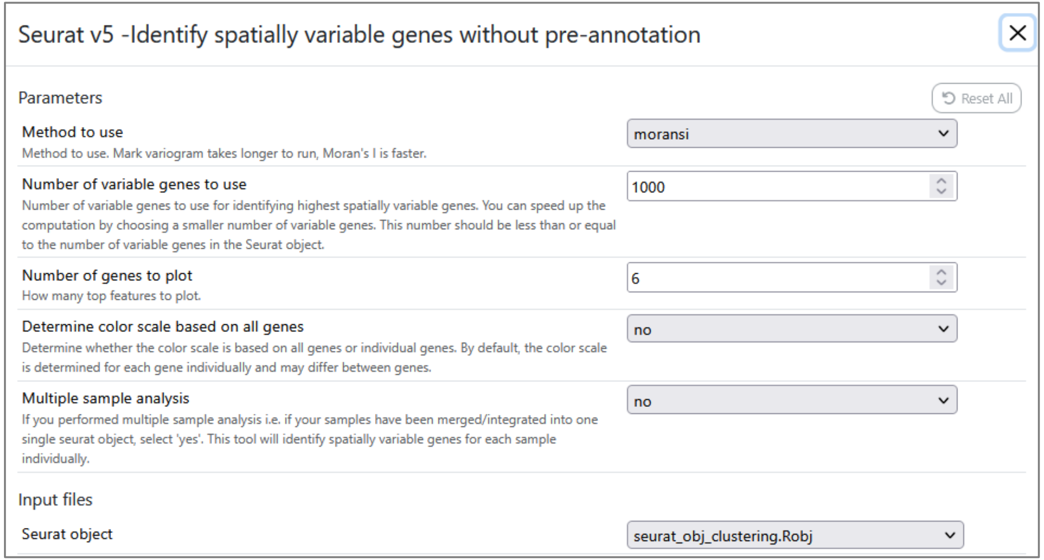
Spreadsheet | Text | Open in New Tab | Details

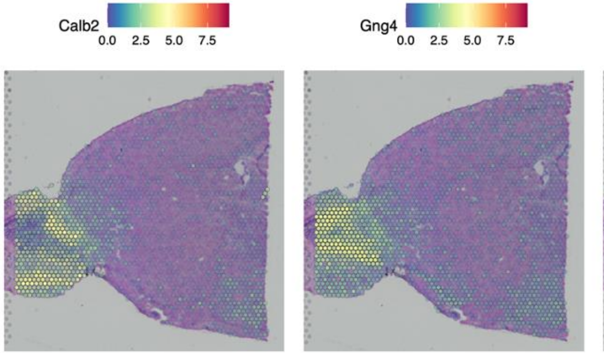
Showing the first 100 of 768 rows. View in full screen to see all rows.

	p_val	avg_log2FC	pct.1	pct.2	p_val_adj
Cplx2	3.92080909952008e-50	1.16425939214283	1	0.99	7.16331822482318e-46
Enc1	8.11425813747733e-49	0.986994243906093	1	1	1.48247496171711e-44
Calb1	1.96410165638653e-48	1.34564614356816	1	0.849	3.58841372621819e-44
Nrsn1	1.97307361691315e-45	-0.904652839123701	1	1	3.60480549810033e-41
Ndfip1	2.83865874686529e-45	-0.653973525431654	1	1	5.18622953052288e-41

Identify spatially variable genes without pre-annotation: Moran’s I

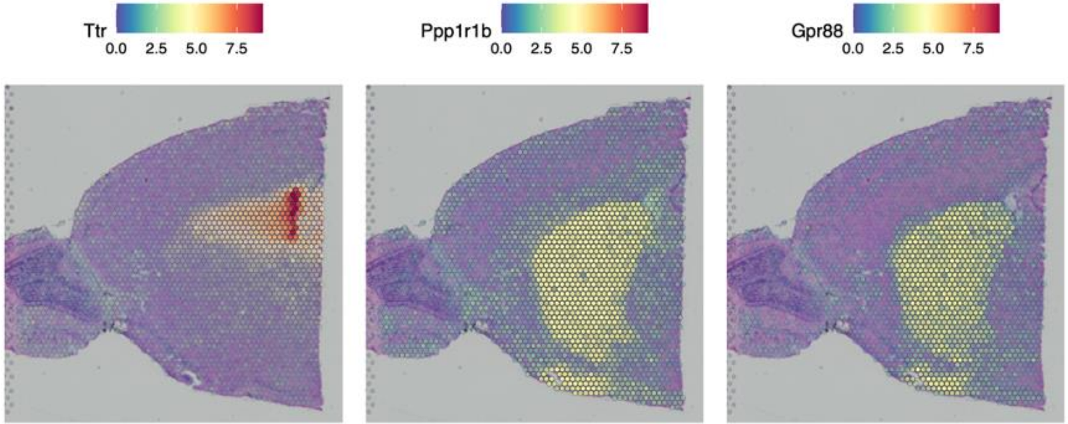
- Measures spatial autocorrelation
  - How much nearby locations resemble each other in terms of gene expression

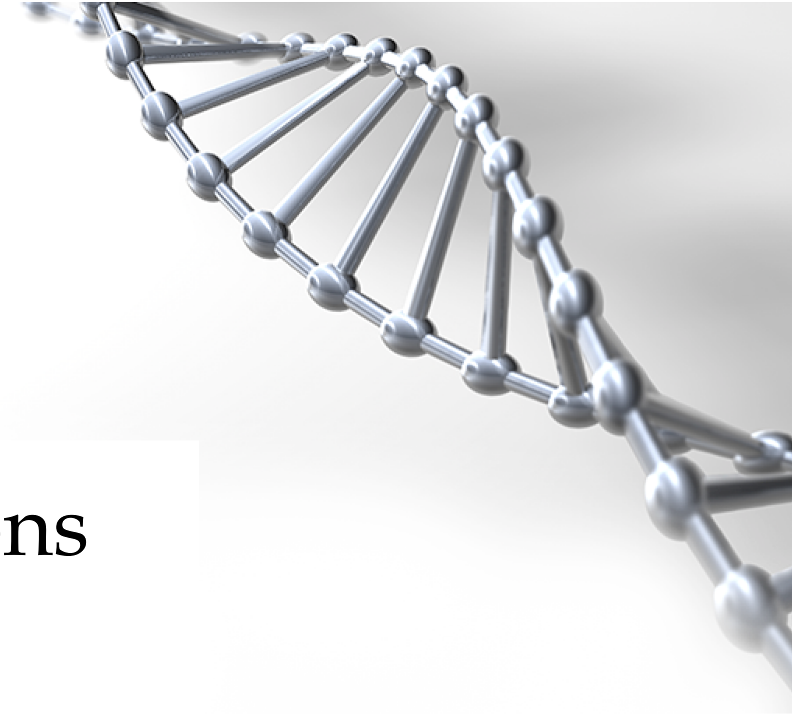




Identify spatially variable genes without pre-annotation: markvariogram

- Models spatial data as a mark point process and computes a variogram
  - Measure of dependence between marks of two spots a certain distance apart
    - By default, Seurat uses a distance of 5, and only computes these values for variable genes to save time.
- Takes a lot of time

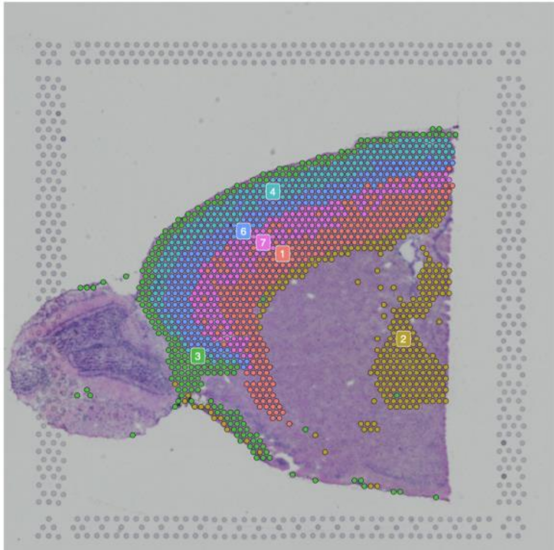




Subset anatomical regions

Subsetting clusters

- Select clusters you want to subset to study them further
  - You can use the “Visualise clusters” tool to more easily see which clusters correspond to specific regions in the slide
- If you are going to integrate with scRNA-seq data, select the clusters that correspond to that data
  - Other clusters can give rise to false positive cell type assignments
  - Note that you need to renormalize and PCA the subset first



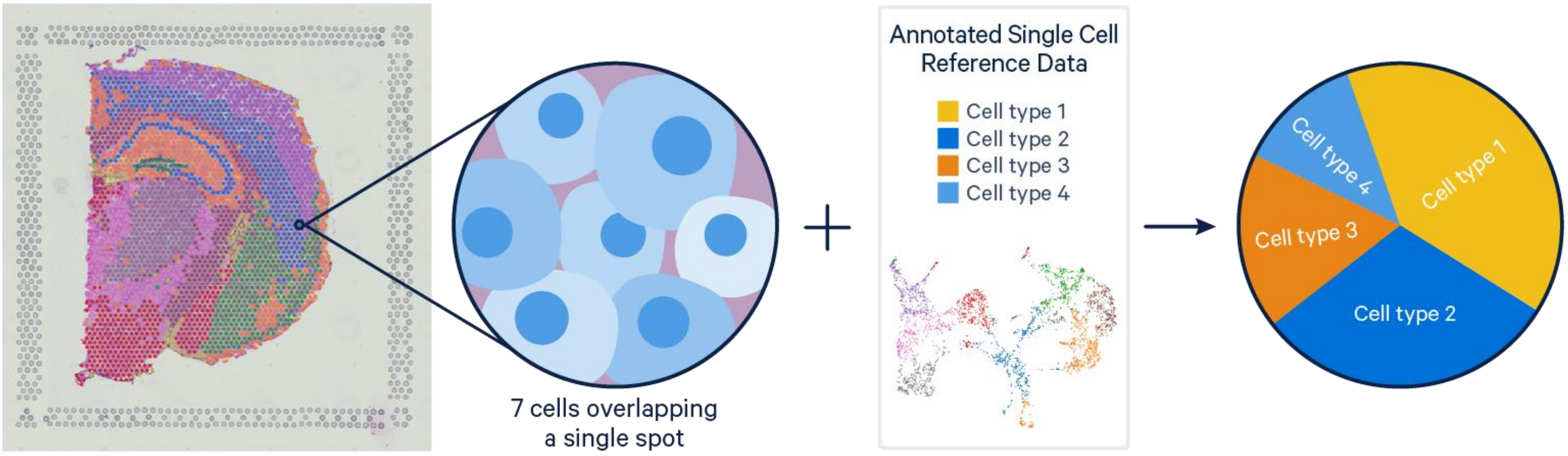
Predict cell type composition of spots: Integrate with scRNA-seq reference data

Spatially resolved transcriptomics

Integration with scRNA-seq reference data

- What is the cell type composition in the spots?
- We use scRNA-seq data as a reference data set
- Several methods available, Seurat uses anchor-based method

Integration with single-cell RNA-seq reference



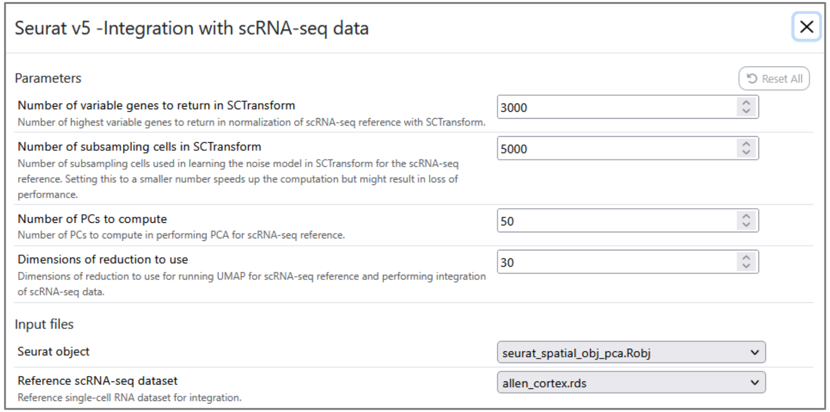
Picture 1. " Integrating Single-cell and Visium Spatial Gene Expression Data" 10xgenomics..com, 17th Mar. 2022, https://www.10xgenomics.com/resources/analysis-guides/integrating-single-cell-and-visium-spatial-gene-expression-data. Accessed 19 Aug. 2022

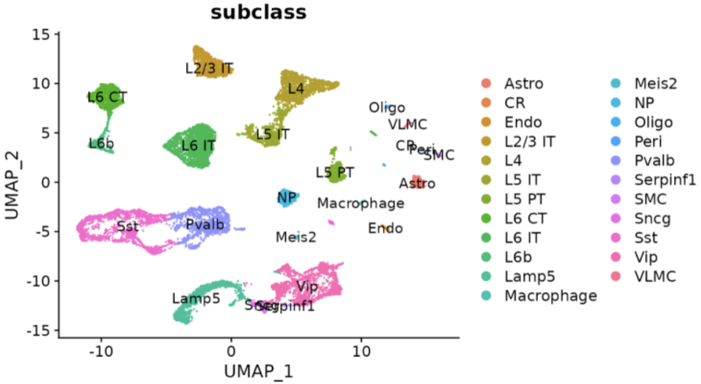
Integration with scRNA-seq data using Seurat

- Find anchors between Visium data and scRNA-seq data (MNN)
- Create correction vector based on differences in expression
- Use correction vectors to remove platform effects
- Integrate data sets
- Transfer cell type information from scRNA-seq data to spatial spots

Seurat v5 -Integration with scRNA-seq data

- First normalizes the reference data set using SCTransform, performs PCA for it, and plots UMAP

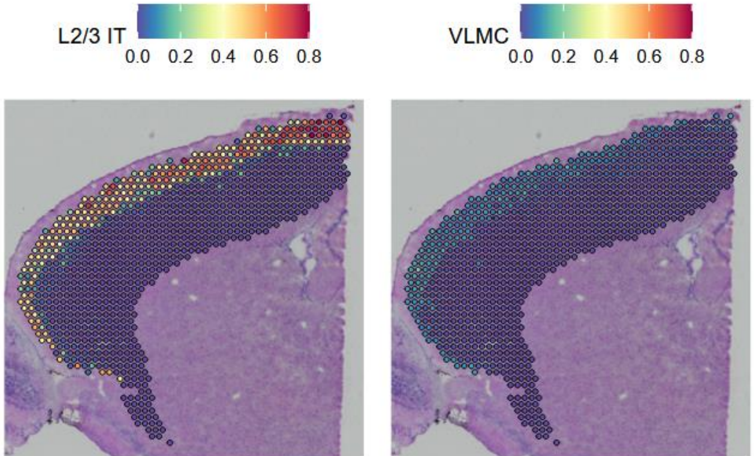


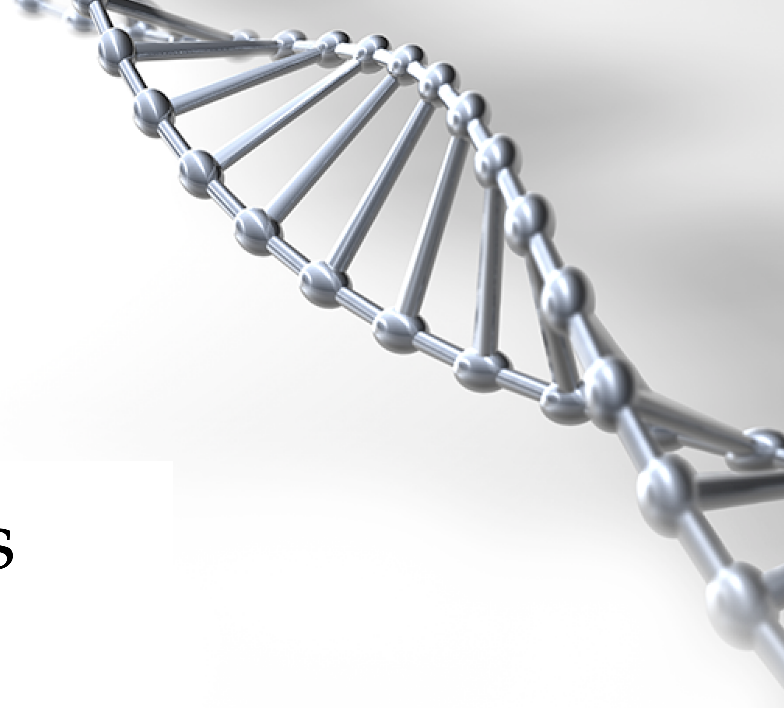


- Reference data should contain all the expected cell types, ideally from the same tissue.
  - it should be a Seurat object stored as an .rds file.
  - cell type annotations should be in the "subclass" column in the metadata slot of the reference object (seurat_obj@meta.data$subclass)
  - gene IDs used should be the same as in the actual data (gene symbols, Ensembl IDs etc)
  - use the tool *Seurat v5 -Extract information from Seurat object* to check your reference data format

Seurat v5 - Visualize cell types after integration with scRNA-seq data

- Visualizes the predicted compositions of selected cell types in each of the spatial spots.
- Also identifies and visualizes predicted cell types whose location is spatially restricted.
  - The same methods are used as to define spatially variable genes (Moran's I and markvariogram), but now cell type prediction scores are used as marks, rather than gene expression
- Input: the Seurat object generated by *Seurat v5 -Integration with scRNA-seq data*.

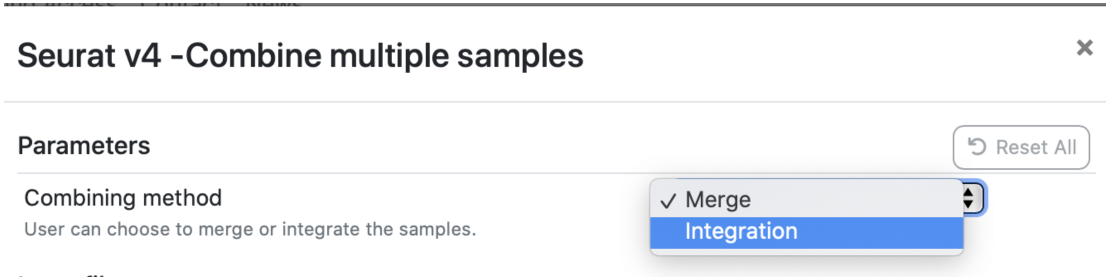


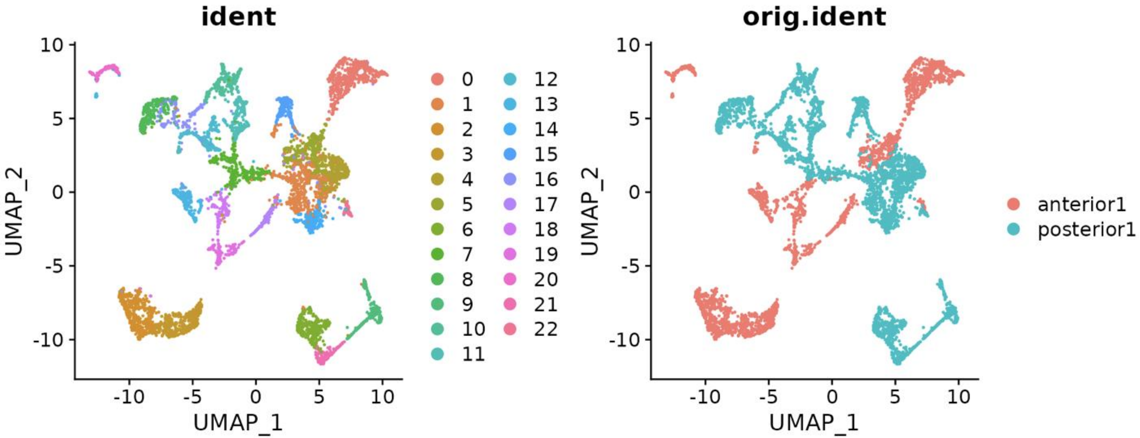


Combine samples

Combine samples

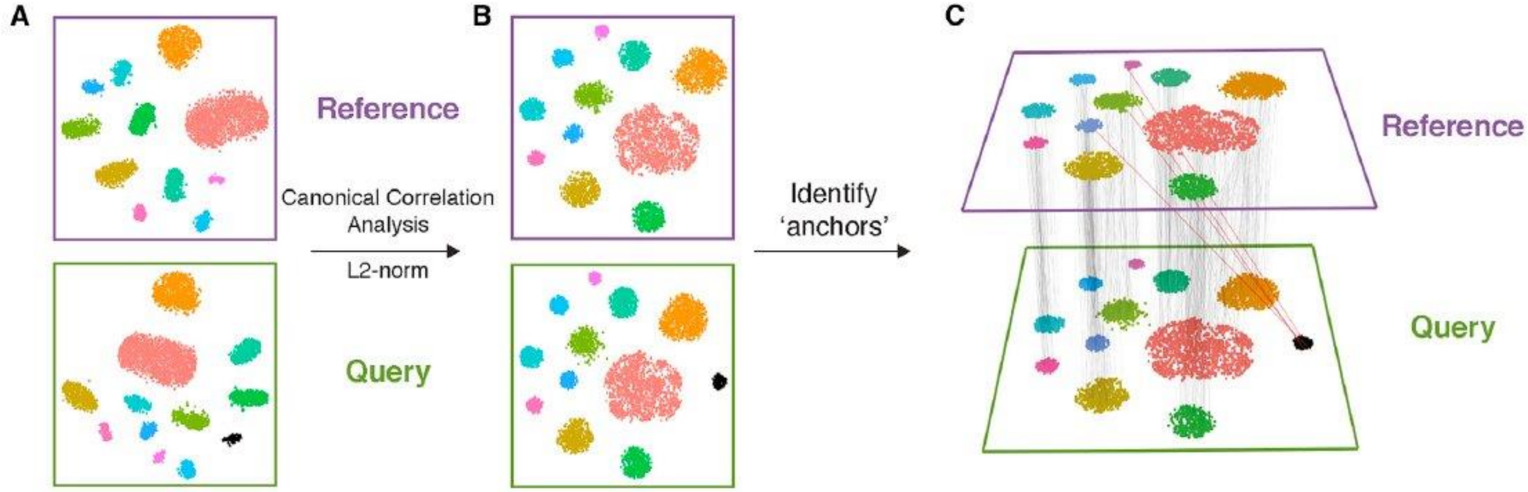
- Currently, two options:
  - Merge
    - Simple 1+1 merging, used in Seurat vignette
    - Ok, when there’s no big batch effect
  - Integrate
    - Similar to what is used to combine samples in scRNA-seq data





Batch effects

- Strong batch effects between samples can affect the multiple sample analysis
  - Can be removed with integration



Picture 1: https://www.nature.com/articles/nbt.4096